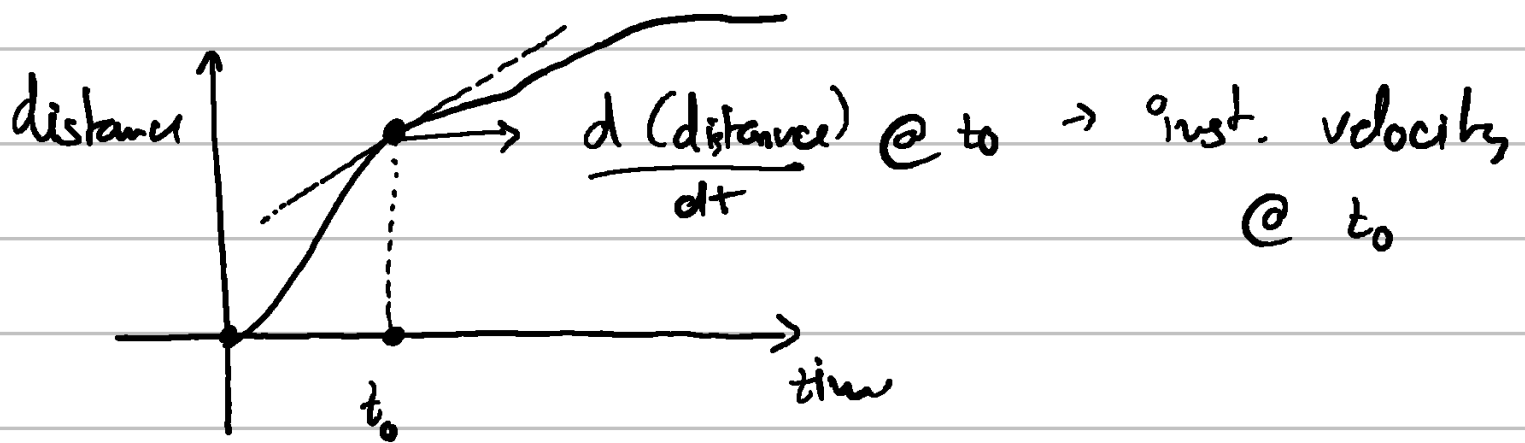


Lesson 1 - Derivatives

Relevance in ML

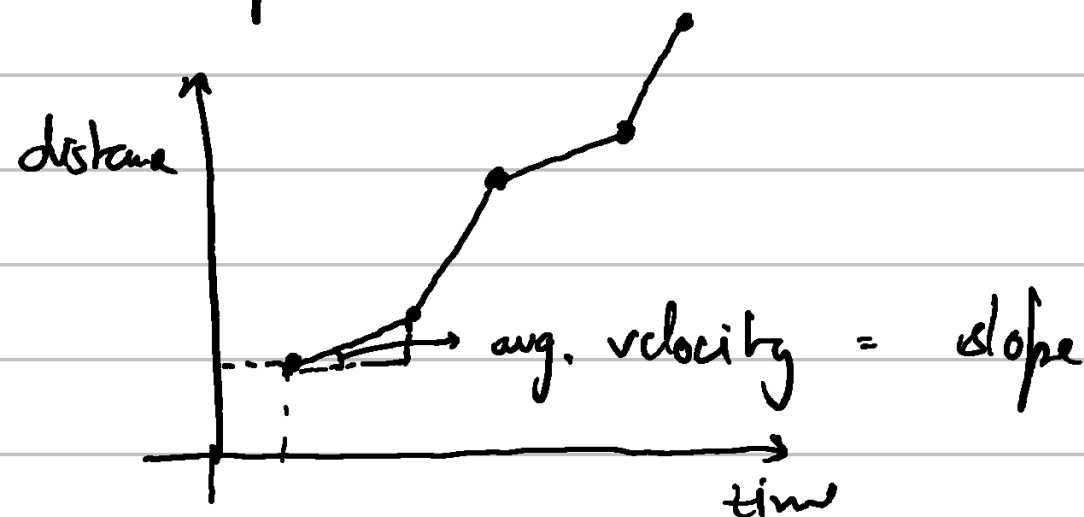
- ↳ Used to optimize mathematical functions.
- ↳ Maximise or minimize.
- ↳ In machine learning, the "best" model is the one whose loss function is minimum.

Derivative $\rightarrow$ The instantaneous rate of change of a function.



$\uparrow$ example of derivative of continuous data.

When data is discrete or in steps, we can only find averages between steps.

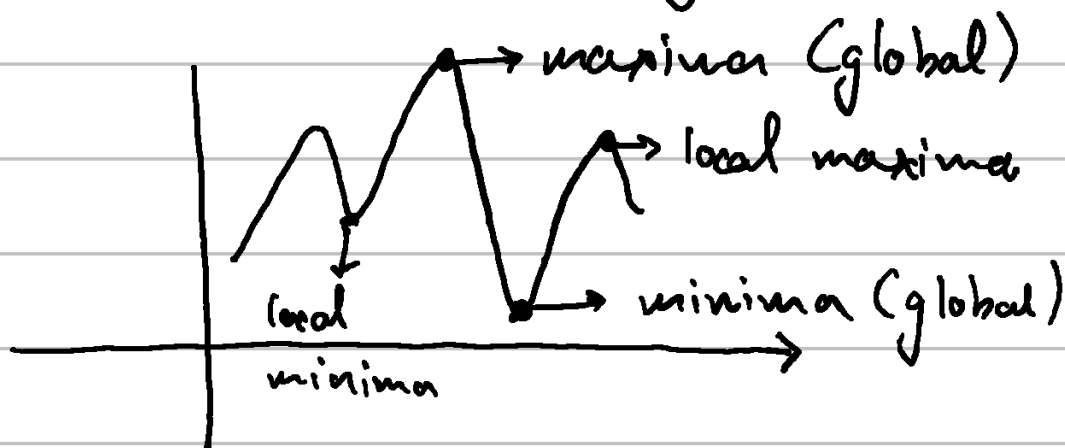


We can keep making our guess better by reducing the steps Δt and as $\Delta t \rightarrow 0$, we get into calculus territory.

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \rightarrow \text{slope of tangent line at point.}$$

Minima $\rightarrow$ The lowest point we have in our data / function

Maxima $\rightarrow$ The highest.



The slope at minima/maxima is zero i.e. the tangent is a line parallel to x -axis.

The vice-versa is true as well, wherever the slope is zero, it is either a maxima or a minima. To find whether it is a maxima or a minima, we check the curvature using second derivative.

$$\Rightarrow \text{slope} = \frac{\Delta y}{\Delta x}$$

$$\text{slope at a point} = \frac{dy}{dx}$$

$$\Rightarrow \text{Lagrange} \rightarrow f'(x)$$

$$\Rightarrow \text{Leibniz} \rightarrow \frac{dy}{dx}$$

Common Derivatives

Polynomials

$$\textcircled{1} \quad y = c$$

$$\frac{dy}{dx} = 0$$

$$\textcircled{2} \quad y = x^n$$

$$\frac{dy}{dx} = n \cdot x^{n-1}$$

$$\textcircled{3} \quad y = a \cdot x^n + c$$

$$\frac{dy}{dx} = a \cdot n \cdot x^{n-1}$$

Inverse of a function is its mirror image using $x=y$ as reference.

$$\frac{\Delta f(x)}{\Delta x}$$

$$\frac{\Delta g(y)}{\Delta y} = \frac{\Delta x}{\Delta f(x)}$$

$$\Rightarrow \frac{d f^{-1}(x)}{dx} = \frac{1}{f'(f^{-1}(x))}$$

$$\textcircled{1} \quad y = \sin x$$

$$\frac{dy}{dx} = \cos x$$

$$\textcircled{2} \quad y = \cos x$$

$$\frac{dy}{dx} = -\sin x$$

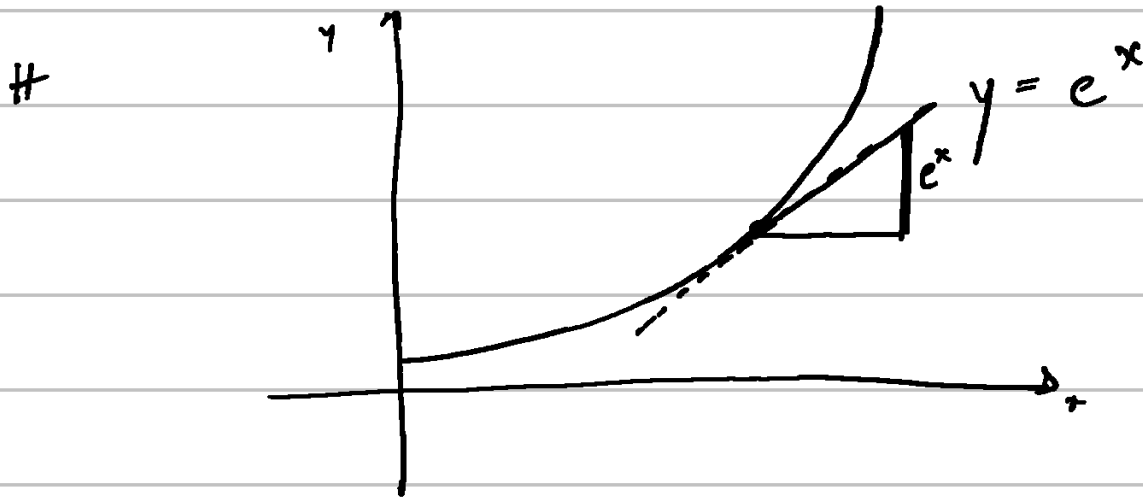
e

$$e = 2.7182 \dots$$

n	1	100	1000
$(1 + \frac{1}{n})^n$	2	2.708	2.717

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

$$\begin{aligned} f(x) &= e^x \\ f'(x) &= e^x \end{aligned}$$



$$y' = e^x$$

#

$$f(x) = e^x$$

$$f^{-1}(x) = \ln(x) = \frac{\log_k x}{\log_k e}$$

⇒ Using this →

$$\frac{df^{-1}(x)}{dx} = \frac{1}{f'(f^{-1}(x))} \quad \text{at } \underline{\underline{\text{point } x_0}}$$

$$f(x) = e^x$$

$$g(y) = \ln(y)$$

$$f'(x) = e^x \longrightarrow g'(y) = \frac{1}{e^x}$$

$$g'(y) = \frac{1}{y} \quad \text{as } \underline{y = e^x}$$

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$$\boxed{\frac{d}{dx} (\ln(x)) = \frac{1}{x}}$$

$\Rightarrow$ Non-differentiable functions $\rightarrow$

- ① Any function with discontinuity at some point.
- ② Functions with a vertical tangent at any point
- ③ Function with abrupt change in values like $f(x) = |x|$ corners/cusps.

Rules

① Sum rule $\rightarrow f(x) = g(x) + h(x)$

$$f'(x) = g'(x) + h'(x)$$

② Product rule $\rightarrow f(x) = g(x) \cdot h(x)$

$$f'(x) = g'(x) \cdot h(x) + g(x) \cdot h'(x)$$

③ Chain rule $\rightarrow f(x) = g(h(t))$

$$\frac{df(x)}{dt} = \frac{dg}{dh} \cdot \frac{dh}{dt}$$

$$f'(x) = g'(h(t)) \cdot h'(t)$$